

Vault-50 Deep Dive – draft

BACKGROUND

Countless replicas of classic British and American amplifiers have been built over the years, and the industry appears to be growing. In the last decade we've even seen more contemporary models from Mesa, Soldano and Friedman "homaged" by third parties.

Typically these are simple electrical copies, and as with anything tone related, results vary. For example the cheeky Ceriatone FAS mimics the Soldano SLO on paper, but the tonal differences are not subtle. Similarly with the dozens of Plexi clones on the market - some are great, others don't capture the Marshall character convincingly at all.



Companies of course take this path because they've realized that most players aren't looking to redefine what great tone is - but are looking for different ways to achieve it. We agree.

These days players and producers are spoiled for choice, whether it be a copy of a classic design, a digital solution or a modern multi-channel amp. We were not interested in exploring the first two categories, but felt like there was much more to do in the latter.

While there have been some excellent offerings in recent decades, we have yet to find one that truly does it all. Once we decided capturing the wall in a single amplifier was an idea worth pursuing, the founding principle was that it cannot be a jack-of-all-trades. It had to provide the closest possible experience to a Zen HQ field trip.

**The innovation behind the Vault-50 isn't DSP – it's
dynamically configurable analog hardware.**

SONIC CLONES

It may seem counter-intuitive, but building an electrical replica of a tube amplifier does not necessarily mean it will nail the sound. In fact, it rarely does.

The many non-linearities in a high-voltage tube device, coupled with wide component tolerances has a very real effect on tone and response. So for each amp model, we began with a reverse-engineered electrical reproduction, and then performed an intensive refinement process to ensure an accurate tonal match with the "clonee".

In concept it's similar to how Friedman approached the Plex amp, and Metropoulos the Metro Plex II. Both builders aimed to sonically clone a specific Plexi of theirs and tweak the circuit to taste.

Our challenge wasn't cloning one great amp. It was fitting an entire amp wall into a single chassis, from a 1964 Fender Pro Reverb to a 2023 Bogner Uberschall Ultra.

THE PROCESS

For each reference amp, comprehensive technical data was collected – including distortion characteristics, frequency response, phase shift, and dynamic behavior. From here an iterative process of incrementally smaller circuit tweaks were made until each gain stage, the tone stack and power amp circuit closely aligned with the source. Initially this match was by the numbers. The listening came later.

This was made possible by exploiting the Vault-50's powerful component-level switching system as outlined on the website. This tech digitally-controls the value and state of every cathode & plate resistor, coupling & bypass capacitor, diode & triode, tonestack configuration, bright cap values – almost everything relevant in the signal path.

Every tiny circuit revision existed for one reason: so changing amp models feels less like selecting a preset and more like walking over to another amplifier.

Some of the methods used for fine-tuning tone and response include:

- altering the level through each gain stage. e.g. via the cathode and/or plate RC configuration.
- shaping the lo and hi frequency roll-off through each stage via cathode, coupling & plate bypass caps.
- changing tonestack resistor and/or capacitor values
- changing voltage dividers (values and ratios) to alter level, shape frequency response and phase shift.
- altering the negative feedback resistor to shape the amp's overall voice and response (looser, rawer etc).
- different output coupling cap values, for a fuller or tighter power section voicing.
- different bright cap values, diode clipping options and more.

Again if the goal is to simply build a simple electrical copy of a well-documented amp, you can paint by numbers and it will sound as it sounds. However once you play through the Vault-50, it will be obvious why we obsessively took the path we did.

Seasoned players constantly asking “which one am I hearing now” is strangely satisfying, yet very telling.

PLAYER TESTS

To ensure we weren't relying on technical data alone, meticulous analysis of recorded samples and blind playing tests with a group of tortured trusted players followed. Some subtle tone tweaks did come from this, but it was mostly the response comments that proved invaluable.

A range of smutty-sounding guitarist words like tight, greasy, hairy, and raw were decoded and the circuits further refined. This process was repeated multiple times if necessary, until identifying the original in blind A/B tests generally came down to guesswork.

It was done old-school, i.e. clone and clonee were set up via an amp-switcher through the same cab, and then toggled back and forth with the player in the room, or control room if they chose.

A few exceptions aside that we will cover soon, an amp model was included only if it elicited a response in one of two basic categories:

1. I can't tell the difference (aka have you switched yet?)
2. I think there's a small difference, but I couldn't tell you which is which.

Any amp that didn't meet either criteria was axed altogether, most notably our Jubilee and Two Rock. Both of these amps have some unique technical aspects that couldn't be accurately replicated in the Vault-50 and although it was a bit of a shame, “it sounds kinda close” was not something we were satisfied with.

MARKETING VS REALITY

It's safe to say many of us are fatigued by over-the-top claims both manufacturers and their fanbase (or paid promoters) make at times. Until recently this was more common in the modeler world but it's increasingly creeping into the analog domain.

Many tales seem to focus around how a tube component or analog device has been replaced with a solid-state, miniaturised or digital version – and always somehow with zero effect on tone. We don't need to name names, but if you pay attention it's starting to resemble the "Kemper can mimic any amp perfectly" talk of 2012.

We tested digital equalizers, Class D power options, small glass and iron but nope – there's no such thing as a free lunch, despite how convenient it may be or how convincingly we are told the code has been cracked. Now we understand that marketing and exaggeration go hand-in-hand, and although our enthusiasm for the Vault-50 may at times smell like it's crossing that line, we are fully-committed to transparency (without giving away the keys to the castle that is).

This is one of the reasons for this document – to provide you with as much information as possible before you potentially make a big purchase.

The simple truth is that the amp models we've created are not identical to the originals. This is a technical impossibility – no two tube amps can ever sound exactly alike. There are too many factors involved, and a tube amp's ecosystem is far too complex for predictable certainty. What we are saying is that the most are extremely close to the originals, and many indistinguishable.

We put these in the 98% club. Calling anything 100% would just be kidding ourselves. and violate our own no-BS terms. Audio clips don't tell the entire story but there will be a video on our YouTube channel soon if not already with some direct comparisons.

A CLOSER LOOK

Some amplifier models recreated in the Vault-50 have operational performance differences to the original amps.

Pot taper

The taper or sweep of different pots makes "putting everything at noon" mean little if comparing the Vault-50 directly with the original. The extremes will sound similar, but along the pot's 300° travel there's generally no simple like-for-like position.

In some cases the OG amp may use log pot in place of a linear one. For example the Bogner Ecstasy treble pot, which typically needs to be set quite high. The Vault-50 uses a linear pot like most other amps, meaning it usually needs to be a few clicks lower to tonally match the Bogner.

In other cases the opposite may apply, so just spin those knobs rather than rely on your old go-to settings. The golden rule – always use your ears and not your eyes to dial in the amp.

Deep Control

The Vault-50 has a deep control that many of the original amps don't. Easy fix – unlike a tone control, the effect of the deep pot is nil when shut off.

It's worth noting though that a number of modern amps (e.g. many Friedmans) don't have a user-adjustable depth knob, but do have a fixed depth circuit under the hood. Running the deep control at 2:00 to 3:00 will recreate the low end bump of those amps.

Options and Switches

The toggle switches found on the original amps (e.g. BE/HBE, FAT, C45 and SAT on the BE100) are generally available via the touch display as a soft switch. However don't be fooled by the digital control aspect - the signal path always remains analog. As is happening in the rest of the amp, physical components are being switched in to authentically recreate the relevant circuit - not simulated via DSP.

The popular options are always available, but in some cases a little fat has been trimmed. For example, the Bogner Ecstasy has more switches than a cockpit, and can be a little overwhelming. We've included the handy Pre-EQ and Boost, but chose to omit the Plexi and Structure options. Similarly for the seldom-used Bass Shift on the Mark series - Bright, Treble Shift and Deep are however included.

As for channels, some like our SLO recreation have all available: Clean, Crunch & Overdrive. For others we've focused on the "one that made the amp famous" - such as the high gain channels on the Wizards and the Mark series (the MkIV does however feature all three). Frankly the lower gain tones on these aren't that special, and better options are abundantly available within the Vault-50.

DEEPER STILL

If you've made it this far you are likely more technically-minded than the average player, so let's flesh out some more underlying variables that "radically" affect tone, or not.

Tube vs Silicon Rectifiers

Most of the amps we've cloned have a silicon rectifier. The notable exceptions are the Pro Reverb, JTM45 and Dirty Shirley. It's fairly subtle, but our versions have a little less sag when being pushed hard. Tonally there is very little difference.

50 vs 100 Watts

Yes the 100W heavyweights in the vault are capable of slightly higher SPLs. On average though it's only around 2dB as the Mercury Magnetics output transformer we've used punches Wizard-level hard. So unless you're into extreme stage volumes, we're confident you'll have more than enough headroom on tap.

Vintage Mojo

If you're firmly and unwaveringly of the opinion that 50 year-old transformers and vintage capacitors are mandatory for great tone, then we're not here to convince you otherwise. For everyone else here's our take. We love old Marshalls. They're fun to play, almost always sound great, and look the part. But as builders such as Metropoulos and Germino have proven time and time again, modern components can absolutely achieve authentic British rock tones - no fairy dust or mojo required.

We've built and owned many Plexi and Master Volume clones that hung with our originals no problem, so we were confident going in that with the right iron, layout and components, the Vault-50 could too. And it does, despite not having gold knobs and a big M logo.

Capacitors: many anecdotal tales of the huge tonal differences between various types and brands have been extolled by internet gurus. We rigorously tested this and unsurprisingly discovered things are far more subtle than the pundits suggest. However we observed a small but helpful difference by implementing a mixture of different caps, namely Synergy Royal Mustards and modern WIMA MKPs.

Output transformer: no the Vault-50 isn't loaded with a myriad of different models that switch in real time. Once again we undertook extensive testing on many types and brands to hear how they stacked up. We confirmed that the difference in performance between a cheap transformer and a quality one is noticeable - no surprises there. However any differences between top-shelf options were inconsequential.

For great all round tone, excellent build quality and pure grunt we ended up with a custom Mercury Magnetics wind, similar to that used by Wizard. It performs extremely well across the board and more than keeps up with a good old Dagnall, Drake or Schumacher for that matter.

THE 95% CLUB

We meant what we said about excluding any amp that didn't fool players in the room, but we think you'll find these exceptions are worthy ones.

Non-Master Volume Adventures

Old 4-input Marshalls are great fun, but impractical for modern players. Most of the crunch comes from the power section distorting – so if you want it dirty, you'll need to play very, very loud.

To overcome this we developed a vintage-voiced saturation circuit, similar to what Metropoulos introduced in their swansong amps. It sits late in the preamp but before the master volume, meaning that neither the power section nor your ears need to be assaulted.

Separating distortion from sheer volume allows users to conveniently set the channel Gain and Volume to taste, much like a JCM800. This circuit is switched in automatically for the JTM45, Plexi, Super Bass and Ed amp, and a modified, American-voiced version is used for the Pro Reverb, Bassman and Buxom Betty. Yes driven Fender tones matter too.

Don't get us wrong – all of these models are very well replicated in the Vault-50. But the dirtier they get, the more they diverge from the originals, and so enter the 95% category. Tonally very close, but the power section near exploding in those old Marshalls has a certain something that requires an external reamping solution to fully harness.

The numerous plexi clones out there with PPIMVs (Friedman, Suhr) and other volume reduction schemes are in the same boat, despite the claims to the contrary.

The Fenders were maybe 97% – even a cranked Blackface Bassman was very tricky to differentiate in the room.

4-Hole course

You may have noticed we opted out of having four inputs and two volumes on our '69 Plexi and '71 Bass clone. Instead a 3-position Channel Blend switch selects either the Bright channel on its own, or a sonic recreation of mixing in the Normal channel a little (~25%), or equally (50%). The Blend feature is available on the Hiwatt, and is a great way to beef up single coils. We didn't bother with the JTM45 as there's plenty of low end on the so-called "High Treble" channel.

Both also feature hi/low inputs and switchable bright caps, and of course the GEQ and Boost are always onhand. More than enough flexibility to dial in a wide variety of classic British tones at any volume.

Diezel Compression

Tonally we got there, but the OG Diezels have a very compressed feel that we haven't fully replicated, i.e. our VH4 and Herbert models are more open and dynamic. A compressor in the loop can mimic the squish if you're missing it, but we've found most players really don't.

The "AC" amp

Covered in the FAQ section, but in case you missed it the circuit we settled on is not particularly close to our original '65 Top Boost. Cathode biased EL84's, the Cut control, that weird tonestack – it was never going to happen. However after some heavy experimenting we ended up with an AC50 meets lean JTM45 Frankencircuit, and it does the job surprisingly well.

The reason we chose to include it as an option of one of the Artist Amps is fairly apparent – it's an homage to classic top boost tone, not a sonic clone.

NON-BELIEVERS

There was some initial skepticism from a few beta-testers about how the Vault-50 can possibly be modeler-like in flexibility, without relying on DSP. Seven years of intensive R&D, that's how.

The only exceptions that aren't really exceptions:

- The Reverb: Yes it's digital. But it's optional, and importantly its 100% wet output is blended into the analog path – your core tone is not digitized. This is known as a “dry-through” design, and completely preserves what's important.
- Using an IR: Digital by nature of course. Again optional, and again does not disrupt regular programming. The signal present at the IR output is tapped off the main signal to the IR block – it affects nothing else in the signal path.

ALTERNATIVES

The only vaguely similar options we've seen that offer a number of tones in the tube domain are:

30 Synergy modules, plus a few power amps. However massive cost, no programmable GEQ, recallable pots or reactive load.

- The Diezel VHX maybe, but it's an analog/digital hybrid, i.e. the signal is necessarily digitized in order to operate. Perhaps a good option if you love the Diezel tone and not much else.
- Hook Wizard – don't know much about these. Looks cool but seems to have a clean & dirty option only as opposed to multiple voices/modes.
- A H&K TriAmp MkIII? Not sure, we've never gelled with any of their amps, and damn they're ugly.
- A wall of amps. Trust us, this one is not exactly convenient or affordable.

Jokes aside, there's nothing we've come across that does what the Vault-50 does – and this is the very reason it was created.

GET IN TOUCH

If you are interested in the Vault-50 and have questions, please reach out via email. If you're ever in Sydney and are keen to hear the Vault-50 vs. the Wall in person, it's worth asking. We may be able to accommodate you depending on studio availability.

We'll close by admitting it has been an incredibly challenging project, but we are proud of what we've accomplished. We feel the Vault-50 is a significant step forward in the historically slow-moving field of tube amplification, and is something we are confident will bring a smile to your face every time you power up.

Disclaimer

All brand names, product names, model names/numbers, artist names, stage names, trademarks, and other identifying marks are the property of their respective owners. Zen Amplification is not affiliated with, sponsored by, endorsed by, or associated with any of these companies, artists, or trademark holders. References to amplifier, speaker, and equipment manufacturers, as well as references to artists and their associated tones, rigs, or musical styles, are used solely for identification and descriptive purposes to indicate tonal inspiration. Such references do not imply any endorsement, sponsorship, approval, or association.